i) Pick and Place Object at Destination

```
robot arm = connect("sim:panda")
task pick_and_place(obj, dest):
     move arm.tool to obj.above(10cm)
     grip arm.gripper close
     move arm.tool to dest.above(10cm) speed 0.2m/s
     grip arm.gripper open
         on fail:
             disp "I couldn't complete that"
pick_and_place(cup, tray)
```

ii) Delivery rover in an office (wheeled)

```
robot rover = connect("office:rover-2")
move rover.base to places.mailroom
disp "Please place the parcel on my tray"
wait until rover.tray.loaded timeout 2min
move rover.base to places.desk_14, speed 0.8m/s
disp "Parcel for you"
wait until not rover.tray.loaded timeout 5min
move rover.base to places.dock
```

iii) Inspection drone (flight profile)

```
robot drone = connect("field:drone-7")
safety:
     when drone.battery < 25%:
          returnhome
takeoff drone(15m)
for point in route.roof_inspect:
    move drone.base to point.at_height(10m)
    capture drone.camera
    save as "roof_{point.id}.jpg"
land drone at places.launch_pad
```

iv) Quadruped patrolling a factory at night

```
robot dog = connect("plant:quad-1")
repeat for 30min:
    for checkpoint in route.night_patrol:
        move dog.base to checkpoint
        if dog.sense("thermal").max > 80cdeg:
            alert "Hot spot at {checkpoint.name}"
            capture dog.camera
            save as "hotspot.jpg"
    move dog.base to places.charger
```

v) Humanoid in a kitchen

```
robot helper = connect("home:humanoid")
move helper.base to places.counter
move helper.right_hand to mug.handle
grip helper.right_hand close
move helper.left_hand to kettle.handle
grip helper.left_hand close

pour from kettle into mug until mug.fill >= 80%
place kettle on counter.spot_a using helper.left_hand
say "Your tea is ready"
```

vi) Industrial welding arm using a vendor extension

```
robot welder = connect("line3:examplecorp-x7")
move welder.tool to seam.start
if welder.has("examplecorp.adaptive_weld"):
    examplecorp.adaptive_weld(seam, current=180A)
else:
    weld along seam speed 5mms current 180A
move welder.tool to places.home
```